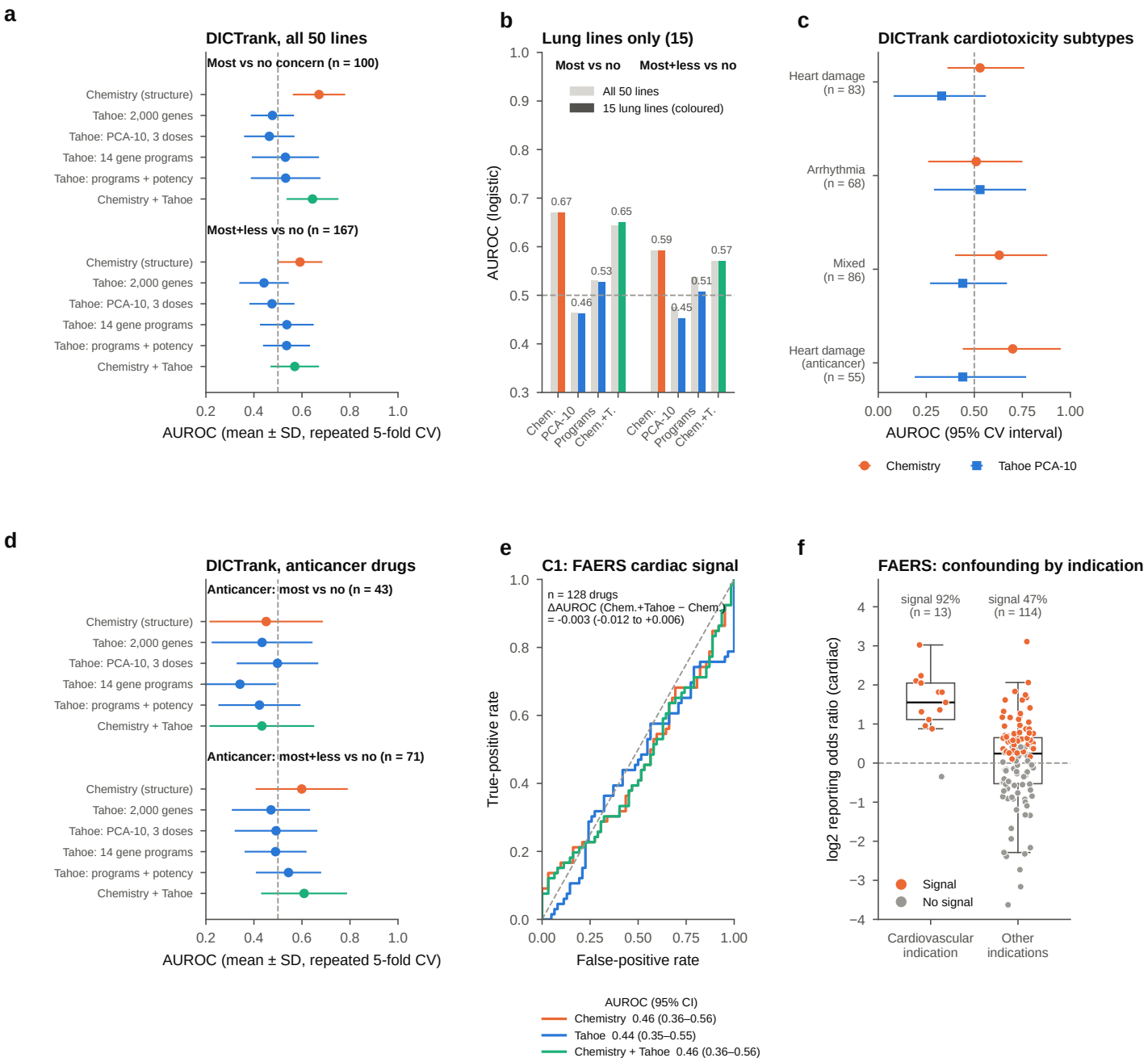
a
DICTrank, all 50 lines
Most vs no concern (n = 100)
Chemistry (structure)
Tahoe: 2,000 genes
Tahoe: PCA-10, 3 doses
Tahoe: 14 gene programs
Tahoe: programs + potency
Chemistry + Tahoe
Most+less vs no (n = 167)
Chemistry (structure)
Tahoe: 2,000 genes
Tahoe: PCA-10, 3 doses
Tahoe: 14 gene programs
Tahoe: programs + potency
Chemistry + Tahoe
0.2
0.4
0.6
0.8
1.0
AUROC (mean ± SD, repeated 5-fold CV)
b
Lung lines only (15)
Most vs no
Most+less vs no
All 50 lines
15 lung lines (coloured)
AUROC (logistic)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.67
0.46
0.53
0.65
0.59
0.45
0.51
0.57
Chem.
PCA-10
Programs
Chem.+T.
Chem.
PCA-10
Programs
Chem.+T.
c
DICTrank cardiotoxicity subtypes
Heart damage (n = 83)
Arrhythmia (n = 68)
Mixed (n = 86)
Heart damage (anticancer) (n = 55)
0.00
0.25
0.50
0.75
1.00
AUROC (95% CV interval)
Chemistry
Tahoe PCA-10
d
DICTrank, anticancer drugs
Anticancer: most vs no (n = 43)
Chemistry (structure)
Tahoe: 2,000 genes
Tahoe: PCA-10, 3 doses
Tahoe: 14 gene programs
Tahoe: programs + potency
Chemistry + Tahoe
Anticancer: most+less vs no (n = 71)
Chemistry (structure)
Tahoe: 2,000 genes
Tahoe: PCA-10, 3 doses
Tahoe: 14 gene programs
Tahoe: programs + potency
Chemistry + Tahoe
0.2
0.4
0.6
0.8
1.0
AUROC (mean ± SD, repeated 5-fold CV)
e
C1: FAERS cardiac signal
n = 128 drugs
ΔAUROC (Chem.+Tahoe − Chem.)
= -0.003 (-0.012 to +0.006)
True-positive rate
1.0
0.8
0.6
0.4
0.2
0.0
0.00
0.25
0.50
0.75
1.00
False-positive rate
AUROC (95% CI)
Chemistry 0.46 (0.36–0.56)
Tahoe 0.44 (0.35–0.55)
Chemistry + Tahoe 0.46 (0.36–0.56)
f
FAERS: confounding by indication
signal 92% (n = 13)
signal 47% (n = 114)
log2 reporting odds ratio (cardiac)
4
3
2
1
0
−1
−2
−3
−4
Signal
No signal
Cardiovascular indication
Other indications